



Outcome, Explainable, and Temporal AI Framework for the Prediction of Social Media Addiction Risk

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ABSTRACT

Social media addiction is a behavioral problem experienced widely today, and usually goes unrecognized until it has reached critical levels. Yet, current research and computational solutions largely assume the problem of addiction detection and prevention to be a static binary classification problem, which is not in line with the actual nature of addiction as a constantly evolving process impacted by various psychological and contextual conditions.

This study proposes an explainable, outcome-driven, and temporal machine learning algorithm for the purpose of predicting social media addiction risk. Our solution is a web-based system, based on behavioral and contextual inputs such as users' level of stress, sleep, moods, and craving. We finalized the XGBoost algorithm for our predictive model, reaching highly competitive results in experiments (89.67% of accuracy).

Moreover, besides the efficiency of our predictive method, the developed system is focused on two important issues of the field: explaining its decisions and incorporating the temporality of addictive processes. Specifically, SHAP (SHapley Additive Explanations) technique is used for explainability, allowing to obtain more reliable predictions, while temporal tracking enables analyzing users' behaviors dynamically.

Keywords: Social Media Addiction, Explainable AI, Temporal Modeling, XGBoost, SHAP, Behavioral Analytics, Outcome-Driven AI

1. Introduction

The emergence of a digital revolution has had an impact on the manner in which people communicate. In the last years, social media platforms have found wide popularity among Internet users due to their ability to quickly transfer information. At the same time, excessive use of social media leads to the occurrence of social media addiction accompanied by high stress levels and bad mood regulation along with low life satisfaction [1], [2].

A commonly used approach to assess addiction is self-report questionnaires based filled out by users themselves. Such a procedure has been shown to produce subjective results, making the approach to using machine learning systems more relevant and efficient [9].

However, addiction prediction models used currently have several drawbacks. First of all, some models assume that addiction can be considered separately from other factors affecting one's behavior. This consideration is incorrect since a constant change in the behavioral characteristics of people makes such models inefficient [2], [11].

Furthermore, there is another limitation to the research in the area, which is associated with the inability of current tools for machine learning to create logical predictions. Although machine learning models are successful in recognizing certain tendencies, they do not provide the reasoning behind them. To solve this problem, different techniques are used, e.g., SHAP [4], [15].

Another drawback in the current addiction models is that they cannot consider various aspects of one's behavior. Since a person's actions are affected by time-related components, it is impossible to estimate the probability of a particular condition being observed by analyzing one of the characteristics of a person.

In connection with what has been mentioned above, the purpose of the current study is to develop an efficient addiction prediction model. An effective algorithm for predicting the probability of addiction can be XGBoost, which works successfully with structured data [16].

2. Literature Review

The social media addiction research conducted within this paper can be categorized into two main areas. These include psychological drivers of social media addiction and development of machine learning models for predicting addiction risk.

Other supportive elements include Architecture itself, Outcome Evaluation Module, which is the process of identifying predictions' accuracy by comparing the results with previously made ones, Temporal Modeling Module, which tracks behavioral changes in time and recognizes trends and patterns, Simulation Module, which gives the option to simulate hypothetical behavioral changes and their influence on future predictions, Data Store Layer, which stores information about the user's details and history of risk, Interactive Dashboard Visualization Layer, which presents the results, Report Generator Module, which generates reports, and finally the Administrative Module, which gives an opportunity to control the system.

This is an approach that guarantees proper functioning of the behavioral surveillance in terms of interpretation and dynamic behavior analysis.

3.2 Data and Features

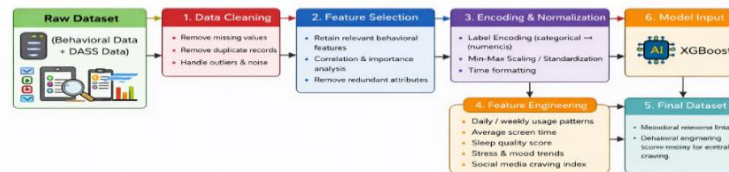


Figure 2 Dataset Workflow Pipeline

The database in this research is obtained by means of questionnaires including common psychological tests such as the Depression, Anxiety, and Stress Scale (DASS). Apart from it, user behavior is measured via daily check-ins that allow analyzing patterns in everyday activity.

The features considered in this analysis can be divided into static and dynamic ones. While static features refer to questionnaire responses, they are related to the overall behavioral pattern of the user and do not change over time.

As for dynamic features, they contain information about such factors as current mood, level of stress, and patterns of user interaction. Dynamic features are relevant in terms of short-term fluctuations.

By considering all of these features, it is possible to assess not only long-term but also short-term patterns of user behavior. After collecting all required features, preprocessing operations are performed including cleaning and normalization of the data. Furthermore, feature selection is required at this stage.

3.3 Model Development

To build a model for predicting the probability of involvement in social media interaction, some machine learning methods were considered.

The methods used included Logistic Regression, Support Vector Machine, Random Forest, and XGBoost. From the experiments conducted, all the methods demonstrated similar effectiveness, but XGBoost proved to be more accurate, giving an accuracy rate of 89.67%. On the other hand, the accuracy rates of the Random Forest, SVM, and Logistic Regression models were 87.57%, 88.62%, and 88.74% respectively.

While there were no considerable differences between the effectiveness of the models used, XGBoost was selected based on its great ability to generalize feature interactions, among others.

3.4 Explainable AI Layer

In order to achieve greater transparency, the explainability module, namely SHAP, has been added into the system. This algorithm makes it possible to determine contributions of particular features to each prediction.

Thus, by analyzing SHAP values, the system will be able to highlight behavioral attributes that are crucial for identifying risky addiction.

3.5 Temporal Tracking

To consider changes in behavioral tendencies, temporal tracking is used in the framework. The proposed system does not rely on single input and predictions; it constantly analyzes user behavior over time.

In this regard, the model can trace behavioral patterns, detect tendencies towards addiction as well as estimate both short- and long-term changes. Thus, the results of addiction risk assessment become more realistic and variable, depending on constant changes.

To ensure the model's accuracy, it's essential to track time, because social media addiction changes over time due to psychological and external factors.

3.6 Behavioral Adjustment Layer

The behavioral adjustment layer allows monitoring user behavior and adjusting to any changes. The model stores prediction results over time, allowing performing longitudinal analysis of behavior and tracing behavioral patterns.

Thus, the system adapts to user behavior changes continuously, which increases its importance concerning users' actual situation. Adaptive behavioral changes make the system closer to reality, which adds to the accuracy of estimates.

3.7 Workflow

On the whole, the system operates using the adaptive workflow. The workflow starts with data collection through user input, followed by preprocessing. After that, machine learning methods are utilized to estimate user behavior and predict risk. Further, the explainability module is applied and behavior changes are tracked.

Behavioral changes simulation and generation of personalized insights through the dashboard are additional capabilities that improve user experience. Results obtained and all information processed is stored in a database. In the end, users receive reports about their behaviors and risks of addiction.

4. Result and Discussion

Table 1 displays the accuracy rate of each of the classifiers as the criteria to evaluate the performance of the models.

Model	Accuracy (%)
XGBoost	89.67
Logistic Regression	88.74
Support Vector Machine	88.62
Random Forest	87.57

Table 1 Comparison of Performance of Machine Learning Models

It is apparent from the results shown in Table 1 that all models have close performance indicators, the highest of which (89.67%) belongs to the XGBoost model. Nonetheless, due to the minor differences between them, the XGBoost model was chosen for further evaluation owing to its reliability and compatibility with the SHAP approach to explainable AI.

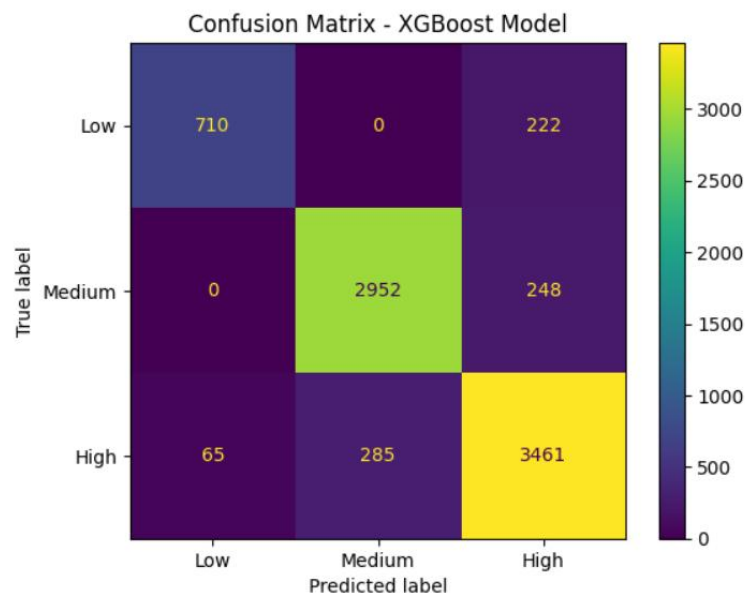


Figure 3 Confusion Matrix of the XGBoost Model

Confusion Matrix of Classifier in Figure 3 demonstrates that the classifier makes balanced predictions of low-, moderate-, and high-risk groups and, therefore, can be regarded as rather effective.

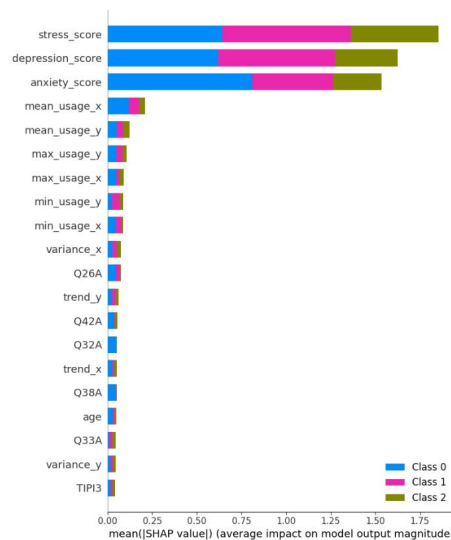


Figure 4 Demonstrates the SHAP Feature Importance Analysis

Figure 4 presents the SHAP feature importance analysis for the model. Based on the outcomes obtained, the following factors contribute significantly to addictive behavior: screen use, nighttime activity, and intensity of behavior, which confirms the findings of other studies [1], [3].



Figure 5 demonstrates the temporal analysis of addiction risk over time

Figure 5 illustrates risk changes at different times. The outcomes obtained suggest that the risks of addictive behavior vary according to time periods and, therefore, should be accounted for when predictions are made [2], [11].

It is evident from the above findings that the proposed framework is successful in merging the elements of prediction, explainability, and temporal analysis in addiction risk prediction, hence making it more realistic and understandable.

5. Conclusion

This study presents an architecture for social media addiction risk prediction using artificial intelligence and machine learning approaches. Through the use of predictive modeling, explainability, and temporal behavioral analysis, this method allows for understanding the risk factors and their dynamics.

The proposed SHAP technique can contribute to enhance the interpretability of models, as it explains the effect of each particular attribute on model predictions. In addition, using temporal modeling, the dynamic analysis of user behavior and predicting the risk of addiction becomes possible. Thus, the above properties of the proposed approach enable overcoming the limitations of previous studies and creating more flexible models.

Our experimental results showed the high performance of the suggested solution, and additionally, new findings regarding the factors contributing to the formation of social media addiction were obtained.

It is possible to continue our research in the future by improving the behavioral data used during the training process for the sake of achieving higher precision.

To conclude, we proposed an effective method for assessing the probability of social media addiction.

References

- [1] N. Çiftci and M. Yıldız, "The relationship between social media addiction, happiness, and life satisfaction in adults: Analysis with machine learning approach," *Int. J. Ment. Health Addiction*, vol. 21, no. 2, pp. 1234–1248, 2023.
- [2] S. Montag, J. Lachmann, M. Herrlich, and K. Zweig, "Addictive features of social media/messenger platforms and freemium games against the background of psychological and economic theories," *Computers in Human Behavior*, vol. 92, pp. 139–147, 2019.
- [3] C. S. Andreassen et al., "The relationship between addictive use of social media and mental health," *Psychology of Addictive Behaviors*, vol. 30, no. 2, pp. 252–262, 2016.
- [4] H. T. T. Nguyen et al., "Evaluation of explainable artificial intelligence: SHAP, LIME, and CAM," in *Proc. FPT AI Conf.*, 2021.
- [5] M. T. Ribeiro, S. Singh, and C. Guestrin, "Why should I trust you? Explaining the predictions of any classifier," in *Proc. ACM SIGKDD*, 2016.
- [6] G. Montag, J. Sindermann, and C. Elhai, "On the psychology of TikTok use: A first glimpse from empirical findings," *Frontiers in Public Health*, vol. 9, Art. no. 641673, 2021.
- [7] M. D. Griffiths and Z. Kuss, "Adolescent social media addiction and mental health," *Int. J. Environ. Res. Public Health*, vol. 14, no. 3, 2017.
- [8] A. G. Kircaburun, M. D. Griffiths, and Z. Demetrovics, "Problematic online behaviors among adolescents and emerging adults," *Journal of Behavioral Addictions*, vol. 7, no. 2, pp. 364–372, 2018.
- [9] E. Chancellor and M. De Choudhury, "Methods in predictive techniques for mental health status on social media," *IEEE Internet Computing*, vol. 24, no. 1, pp. 24–33, 2020.
- [10] H. Sarker, A. Tyburski, K. Rahman, K. Hovsepian, and S. Kumar, "Finding significant stress episodes in a discontinuous time series of rapidly varying mobile sensor data," *Proc. ACM IMWUT*, vol. 1, no. 3, 2017.
- [11] Y. Wang, M. Chen, and X. Li, "A deep learning approach for detecting social media addiction behaviors," *IEEE Transactions on Computational Social Systems*, vol. 7, no. 5, pp. 1205–1215, 2020.
- [12] S. Orben and A. K. Przybylski, "The association between adolescent well-being and digital technology use," *Nature Human Behaviour*, vol. 3, pp. 173–182, 2019.
- [13] J. M. Twenge, "More time on technology, less happiness? Associations between digital-media use and psychological well-being," *Preventive Medicine Reports*, vol. 12, pp. 271–283, 2018.
- [14] A. Burnap and M. L. Williams, "Cyber hate speech on Twitter: An application of machine classification and statistical modeling for policy and decision making," *Policy & Internet*, vol. 7, no. 2, pp. 223–242, 2015.
- [15] S. M. Lundberg and S. I. Lee, "A unified approach to interpreting model predictions," in *Proc. NeurIPS*, 2017.
- [16] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. ACM SIGKDD*, 2016.
- [17] D. J. Kuss and M. D. Griffiths, "Online social networking and addiction—A review of the psychological literature," *Int. J. Environ. Res. Public Health*, vol. 8, no. 9, pp. 3528–3552, 2011.
- [18] C. S. Andreassen, "Online social network site addiction: A comprehensive review," *Curr. Addict. Rep.*, vol. 2, no. 2, pp. 175–184, 2015.
- [19] J. Bollen, H. Mao, and X. Zeng, "Twitter mood predicts the stock market," *J. Comput. Sci.*, vol. 2, no. 1, pp. 1–8, 2011.
- [20] M. Montag and C. Walla, "Carrying out research in the digital age: New perspectives on smartphone addiction," *Addict. Behav. Rep.*, vol. 4, pp. 1–5, 2016.
- [21] E. M. B. Bányai et al., "Problematic social media use: Results from a large-scale nationally representative adolescent sample," *PLoS ONE*, vol. 12, no. 1, 2017.
- [22] S. Shensa et al., "Social media use and depression and anxiety symptoms: A cluster analysis," *Am. J. Health Behav.*, vol. 42, no. 2, pp. 116–128, 2018.
- [23] E. B. Falk et al., "Self-related processing in social media: A behavioral and neural perspective," *Nat. Commun.*, vol. 6, 2015.
- [24] K. Tandoc, P. Ferrucci, and M. Duffy, "Facebook use, envy, and depression among college students," *Comput. Human Behav.*, vol. 43, pp. 139–146, 2015.

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- [25] G. Turel and A. Serenko, "The benefits and dangers of enjoyment with social networking websites," *Eur. J. Inf. Syst.*, vol. 21, no. 5, pp. 512–528, 2012.
- [26] Y. Kwon et al., "Development and validation of a smartphone addiction scale (SAS)," *PLoS ONE*, vol. 8, no. 2, 2013.
- [27] M. T. Ahmed, M. R. Islam, and S. Kabir, "A machine learning approach to detect social media addiction using behavioral data," *Procedia Computer Science*, vol. 152, pp. 319–326, 2019.
- [28] R. A. Davis, "A cognitive-behavioral model of pathological Internet use," *Computers in Human Behavior*, vol. 17, no. 2, pp. 187–195, 2001.
- [29] M. D. Griffiths, "A 'components' model of addiction within a biopsychosocial framework," *Journal of Substance Use*, vol. 10, no. 4, pp. 191–197, 2005.